

Sairam S

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Education

PES University

2023 – 2027

- B.Tech in Computer Science and Engineering (AI&ML)

8.6/10

Publications

The Final-Stage Bottleneck: A Systematic Dissection of the R-Learner for Network Causal Inference

Sairam S, Sara G, Shivam S

Transactions on Machine Learning Research (TMLR), 2026

 |  | W&B | OpenReview

Research Experience

Research Intern

June 2026 – August 2026

WSAI, Indian Institute of Technology Madras | Advisor: [Prof. Krishna Pillutla](#)

- Developing a differentially private LLM framework tailored for private synthetic data generation.

Interests: Differential Privacy, Robustness, Adversarial ML.

Projects

The Geometry of Bit-Collapse: Quantization Instability

Winter '26

- Investigated geometric failure modes of Post-Training Quantization (PTQ) on ResNet-18/CIFAR-10 using a custom **Hessian Spectrum Visualizer** (PyHessian, averaged over 40 batches for variance reduction).
- Diagnosed “**Bit-Collapse**”: Int8 quantization sharply narrows the loss basin and triggers spectral eigenvalue spikes, destabilizing standard SGD training regimes.
- Demonstrated that **SAM** fully restores geometric stability under quantization, recovering a flatter loss basin than the FP32 baseline — suggesting SAM as a principled mitigation for quantization-induced sharpness.

Reproducibility Study: The SAM Optimizer | W&B

Fall '25

- Implemented a clean-room version of **Sharpness-Aware Minimization** (ICLR 2021) from first principles, manually engineering the dual-step perturbation-ascent / gradient-descent logic via explicit gradient normalization.
- Reproduced Wide-ResNet/CIFAR-10 accuracy ($\sim 96\%$ vs. paper's $\sim 97\%$), then identified a structural gap in the original formulation: SAM implicitly outsources model-complexity regularization to weight decay, leaving it unaccounted for in the perturbation objective.
- Characterized a failure regime: the second-order gradient approximation degrades precisely at sharp minima — the regime SAM is designed to escape — revealing a theoretical tension in the algorithm's core assumption.

A First-Principles Dissection of Optimization Pathologies

Fall '25

- Built a NumPy-based **Hessian geometry engine** to compute and visualize 2D loss landscapes (Rosenbrock, Beale), diagnosing the oscillatory failure mode of Vanilla SGD in high-curvature valleys.
- Analytically demonstrated how Momentum dampens gradient noise via exponential averaging, accelerating convergence in ill-conditioned optimization basins.

Reverse-Mode AutoDiff Engine

Summer '25

- Architected a scalar-valued **autograd engine** from scratch, implementing reverse-mode automatic differentiation via local gradient accumulation on dynamically constructed DAGs with topological sort.
- Verified numerical correctness against PyTorch to $< 10^{-6}$ precision across composite differentiable functions, building ground-up intuition for backpropagation mechanics.

Achievements

- Dr. MR Doreswamy Merit Scholarship Awardee

2024

- CN Rao Merit Scholarship Awardee

2024

Talks & Presentations

- Lightning Talks WSAI, IIT - Madras. 2026

Co-Curriculars & Volunteering

- Teaching Assistant - Mathematics for Computer Science - Prof. Anitha.N PES University. 2025